

LYNEHAM, United Kingdom

WMO: 037400

Lat: 51.5031N

Lon: 1.9922W

Elev: 145

StdP: 99.60

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.3	-6.2	2.3	-1.6	-4.9	2.5	-0.6	13.5	7.7	12.2	7.3	3.8	30	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	25.9	18.1	23.9	17.1	22.1	16.3	19.0	23.9	18.0	22.1	17.1	20.7	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	12.6	20.5	16.4	11.9	19.3	15.7	11.3	18.4	54.3	23.9	51.2	22.2	48.5	20.7	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.7	(c) 8.6	(e) -5.9	(f) 29.0	(g) 1.8	(h) 2.2	(i) -7.2	(j) 30.6	(k) -8.3	(l) 31.9	(m) -9.3	(n) 33.1	(o) -10.6	(p) 34.7		
(4) 11.0	(5) 9.7	(6) 8.6	(7) -5.9	(8) 29.0	(9) 1.8	(10) 2.2	(11) -7.2	(12) 30.6	(13) -8.3	(14) 31.9	(15) -9.3	(16) 33.1	(17) -10.6	(18) 34.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.4	4.7	6.4	8.6	11.7	14.6	16.8	16.4	14.1	11.0	7.3	4.9
	DBStd	5.28	3.25	3.13	2.86	2.87	2.86	2.51	2.50	2.37	2.27	2.73	3.05	3.61
	HDD10.0	779	174	148	114	58	14	0	0	0	1	21	90	160
	HDD18.3	3032	432	381	369	291	207	115	63	70	129	228	332	416
	CDD10.0	821	1	1	3	17	66	139	210	199	123	51	8	2
	CDD18.3	31	0	0	0	0	1	4	14	11	1	0	0	0
	CDH23.3	233	0	0	0	1	4	37	113	70	7	1	0	0
	CDH26.7	37	0	0	0	0	0	5	19	13	0	0	0	0
Wind	WSAvg	4.3	4.9	4.8	4.8	4.4	4.4	4.0	3.8	3.7	3.9	4.2	4.3	4.6
Precipitation	PrecAvg	715	68	49	54	51	55	56	50	58	63	67	69	75
	PrecMax	988	132	140	149	153	139	145	117	134	142	155	142	149
	PrecMin	478	7	4	2	1	5	3	5	2	9	5	26	16
	PrecStd	108	35	33	29	29	30	37	27	36	37	38	32	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.8	16.2	21.0	23.8	27.1	28.9	28.4	24.4	19.5	14.8	13.1
		MCWB	11.1	10.2	10.8	14.2	16.2	18.2	18.7	18.6	17.5	16.0	13.4	11.9
	2%	DB	11.2	11.0	13.6	17.5	21.4	23.9	26.4	25.2	21.5	17.1	13.6	12.1
		MCWB	10.2	9.3	9.8	12.1	15.2	17.3	18.1	18.1	16.3	14.5	12.5	11.3
	5%	DB	10.1	10.1	12.0	15.2	19.0	21.8	24.1	22.7	19.6	16.0	12.7	11.2
		MCWB	9.2	8.8	9.2	10.7	14.1	16.0	17.1	16.9	15.2	13.9	11.6	10.3
	10%	DB	9.2	9.2	10.7	13.4	16.8	19.7	22.1	20.8	18.2	15.0	11.8	10.2
		MCWB	8.2	8.0	8.6	9.8	12.8	14.8	16.4	16.1	14.6	13.3	10.8	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.0	11.7	14.8	17.6	20.0	20.1	20.4	18.7	16.6	14.0	12.2
		MCDB	11.9	12.2	14.1	19.3	22.1	25.3	26.3	25.9	22.4	18.8	14.4	12.6
	2%	WB	10.4	9.9	10.7	12.7	15.8	17.8	18.8	18.9	17.2	15.3	12.7	11.5
		MCDB	11.0	10.7	12.6	16.6	19.7	22.2	24.3	22.9	20.0	16.5	13.4	12.0
	5%	WB	9.3	9.1	9.9	11.4	14.6	16.6	17.9	17.8	16.2	14.5	11.8	10.6
		MCDB	10.0	9.9	11.4	14.3	18.2	20.5	22.7	21.1	18.5	15.5	12.6	11.2
	10%	WB	8.3	8.2	9.1	10.4	13.4	15.5	17.0	16.9	15.3	13.6	10.9	9.4
		MCDB	9.0	9.1	10.5	12.9	16.4	19.0	20.8	19.9	17.4	14.7	11.7	10.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	6.6	7.9	8.1	8.3	8.6	8.0	7.6	6.2	5.2	4.7
		MCDBR	5.5	5.7	8.3	10.5	11.3	11.8	12.3	10.9	9.4	6.6	5.2	5.0
		MCWBR	5.3	4.7	5.4	6.2	6.3	6.2	5.5	5.1	5.0	4.5	4.5	5.0
	5% WB	MCDBR	5.5	5.5	7.2	9.3	10.1	10.2	10.5	9.1	8.2	6.1	5.2	5.0
		MCWBR	5.5	5.0	5.3	5.9	6.0	6.0	5.6	5.0	5.2	4.8	4.8	5.2
Clear-Sky Solar Irradiance	taub	0.321	0.336	0.370	0.391	0.395	0.390	0.386	0.376	0.362	0.348	0.322	0.311	
	taud	2.363	2.381	2.285	2.239	2.262	2.304	2.322	2.373	2.404	2.425	2.411	2.341	
	Ebn at Noon	685	784	819	844	858	865	861	853	822	759	681	629	
	Edh at Noon	63	80	106	124	128	123	119	108	94	76	59	55	
All-Sky Solar Radiation	RadAvg	0.80	1.47	2.58	3.96	4.84	5.31	5.00	4.14	3.12	1.79	0.95	0.62	
	RadStd	0.07	0.19	0.34	0.46	0.28	0.56	0.49	0.35	0.25	0.16	0.10	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page